c. Connectivity

Another factor affecting the number of competing consolidators is the uncertainty regarding connectivity charges for data underlying consolidated market data and their effects on the viability of the decentralized model. Each exchange's data connectivity fees will continue to be set forth in the exchange's fee schedules and must continue to meet statutory standards.[2062] Connectivity fees for the provision of data content underlying consolidated market data would be a fixed input cost for competing consolidators, and, therefore, the level of connectivity fees for data content underlying consolidated market data may affect the economies of scale and the resulting number of competing consolidators.[2063] To the extent that some competing consolidators choose to offer data products with narrower data content than the entirety of consolidated market data, they could lower their connectivity costs because they could likely use connectivity options with narrower data transmission bandwidths.

d. Potential Size of the Market for Consolidated Market Data Products

Another important factor in assessing whether competing consolidators might face profitable business opportunities is the size of the market for consolidation and dissemination services. The size of the market will limit the aggregate revenue that competing consolidators

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2062 See Proposing Release, 85 FR at n.1019.

2063 One commenter stated that the Commission may need to consider ways "to avoid the imposition of fees that are substantially disproportionate to the cost of providing these connectivity methods." See IEX Letter at 8. As discussed below, connectivity fees competing consolidators might pay to the exchanges to receive data content underlying consolidated market data will have to be filed with the Commission as part their fee schedules and must continue to meet statutory standards. See infra Section V.C.2(b)(i)c and note 2171.

will be able to collect from market participants. The size of the market can only support the number of competing consolidators that keep aggregate costs at or below the aggregate revenue.

Commenters stated that the size of this market is not large enough to support enough competing consolidators for sufficient competition and that the Proposing Release did not adequately analyze potential revenue streams for competing consolidators.[2064] The Commission believes that the size of the market is large enough to sustain several competing consolidators, because the Commission estimates that the potential annual revenues for competing consolidators will range from approximately $78 million to $97 million.[2065] This is large enough to support several competing consolidators. The Commission is able to estimate the current revenues from consolidation and dissemination of SIP data because of some new information provided by one commenter.[2066]

The Commission believes these estimates are a lower bound[2067] and are based on the current SIP market conditions. They do not take into account any demand expansion from potential new entrants into the broker-dealer, market maker, and other latency sensitive

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2064 See, e.g., IDS Letter I at 3.

2065 To calculate these numbers the Commission uses estimates of the current revenues from consolidation and dissemination of SIP data as well as estimates of potential revenues from market participants switching from proprietary data to consolidated market data products as a proxy for the potential revenue size for the new competing consolidator business. See infra notes 2072, 2074, 2075, 2076, and 2077 for the calculations of these numbers and the various assumptions that went into those calculations.

2066 See Nasdaq Letter IV at 29 for a discussion on Nasdaq's connectivity and market data revenue numbers. See also Proposing Release, 85 FR at 16816.

2067 These are estimates for the end of 2018, because the main connectivity information used in these calculations is provided by one of the commenters for 2018. See Nasdaq Latter IV at 29.

businesses[2068] nor from market participants who currently rely exclusively on SIP data choosing to spend more on data to receive additional consolidated data. [2069] The Commission cannot address these omissions because it does not have sufficient information to estimate the size of this potential demand expansion. As a result, these numbers underestimate the potential market size for competing consolidators. In addition, the estimates contained in this section are associated with significant additional uncertainty, especially in terms of connectivity revenues.[2070] The potential revenue estimate is based on the current exclusive SIPs' revenues combined with certain market data aggregators' and certain exchanges' revenues that the Commission believes could be available for competing consolidators under the amendments. Specifically, the four components of these estimated potential revenues are: current exclusive SIP operating expenses (approximately $16 million), fees paid to the current SIP data normalizers (approximately $21 million), SIP data connectivity fees paid to the exchanges

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2068 See infra Section V.C.4(b) for a discussion on potential new entrants into the broker-dealer, market maker, and other latency sensitive businesses.

2069 Potential demand for consolidated market data under the amendments is unlikely to be smaller than the current demand from market participants who rely on SIP data because market participants will continue to need the NBBO and last sale information to comply with the Vendor Display Rule. The Commission believes that the potential demand for consolidated market data might be larger than the current demand from market participants who rely on SIP data, because a portion of the current proprietary data users might switch to using consolidated market data and, additionally, there might be new entry into the broker-dealer, market maker, or other latency sensitive businesses, as discussed below in Section V.C.4(b). One of the commenters agreed that some of the current proprietary data users might switch to using consolidated market data. According to the commenter, a portion of those market participants newly choosing to use consolidated market data could become self-aggregators and others could be served by competing consolidators. See Nasdaq Letter IV at 25. See also supra Section V.C.1(c) for a discussion on the benefits of expanded core data content.

2070 See supra note 2065 for additional caveats.

operating the exclusive SIPs (approximately $13 million to $18 million), and data processing and connectivity fees (approximately $28 million to $42 million) from proprietary data users switching to using consolidated market data products.

The first component of the estimated potential revenues for competing consolidators is the operating expenses the current exclusive SIPs collect for their consolidation and dissemination services. The SIP data consolidation and dissemination fees currently paid to the exclusive SIPs could be paid to competing consolidators under the Rule, and thus, could be a potential source of income for the new competing consolidator business. As discussed above, UTP operating expenses totaled around $7 million in 2017 and CTA operating expenses totaled around $8.8 million in 2018.[2071] Together, the Commission estimates the exclusive SIPs' operating expenses to be approximately $16 million at the end of 2018.[2072]

The second component of the estimated potential revenues for competing consolidators is the overall fees market participants pay to market data aggregators that are SIP data normalizers.

2071 See supra Section V.B.2(d) for a discussion on the current exclusive SIP's operating expenses.

2072 The Commission estimates the UTP operating expenses to be approximately $7.4 million as of the end of 2018, based on an estimated 6% rate of increase. This rate of increase is calculated as the change of information services revenues Nasdaq reported in its 2018 and 2019 Form 1 filings and the Commission assumes that a similar rate of increase applies to Nasdaq's SIP operating expenses. Nasdaq's Form 1 filings describe that its market data revenues (excluding connectivity revenues), including from SIP data, are recorded under the information services item of its consolidated income statement. According to its 2018 and 2019 Form 1 filings, Nasdaq's information services revenues increased from approximately $230 million at the end of 2017 to approximately $243 million at the end of 2018, an approximately 6% increase. See 2018 Nasdaq Form 1 filing, available at https://www.sec.gov/Archives/edgar/vprr/1800/18002770.pdf (last accessed Sept. 7, 2020); 2019 Nasdaq Form 1 filing, available at https://www.sec.gov/Archives/edgar/vprr/1900/19003684.pdf (last accessed Sept. 7, 2020).

Current SIP data normalizers take in raw data provided by the two exclusive SIPs and create a combined single data feed to their subscribers. Under the Rule, with the cessation of the exclusive SIPs, these subscribers will likely purchase consolidated market data products from competing consolidators.[2073] The fees that market participants currently pay to the SIP data normalizers might be comparable to what market participants could pay to competing consolidators under the amendments, and thus could be another potential source of income for the new competing consolidator business. Based on its knowledge and expertise, the Commission believes that current SIP data normalizers operate on a price schedule where they charge 6% over the current SIP data fees. This pricing schedule and 2018 total SIP data fees indicate an estimated potential revenue of approximately $21 million to be available for competing consolidators.[2074]

2073 One commenter stated that "[t]he primary ability needed to act as a self-aggregator is technical skill." See AHSAT Letter at 3. The Commission believes that it is very unlikely for the market participants that currently receive data from SIP data normalizers to choose to become a self-aggregator under the amendments given the substantial investment and costs needed to become a self-aggregator. Thus, under the amendments, these market participants will likely purchase their market data from competing consolidators, and not self-aggregate. See supra Section V.B.2(c) for a discussion of the different levels of technical expertise and sophistication market participants have. See supra Section V.B.2(b) for a discussion of SIP data normalizers and their subscribers.

2074 For this estimation, the Commission is using the publicly available SIP revenue information. The total SIP data revenues in 2018 were approximately $164 million for Tape A, $94 million for Tape B, and $132 million for the UTP SIP. Of these revenues, 10% for Tape A, 9% for Tape B, and 12% for UTP were revenues from non-display users. The Commission believes that market participants who purchase data from SIP data normalizers are unlikely to be non-display users, thus the SIP revenues from non-display users should be excluded from this calculation. In 2018, the total SIP data revenues without non-display users is approximately $349 million. The 6% margin over these data revenues will indicate an approximately $21 million potential annual revenue that might be available for competing consolidators. See infra note 2191 for the CTA and UTP Plans' Q2 2020 Quarterly Revenue Disclosures.

The third component of the estimated potential revenues for competing consolidators is the current SIP data connectivity fees paid to the exchanges operating the exclusive SIPs. As discussed above, the exchanges operating the exclusive SIPs charge connectivity fees to SIP data users who directly connect to the exchanges to receive SIP data. Under the Rule, market participants who will use consolidated market data products and who will not self-aggregate will likely pay connectivity fees to competing consolidators instead of the exchanges. The SIP data connectivity fees that market participants currently pay to the exchanges might be paid to competing consolidators under the amendments, and thus, could be another potential source of revenue for the new competing consolidator business.[2075] The Commission estimates that the

[2075] Today the exchanges operating the exclusive SIPs offer connectivity products that bundle SIP connectivity with other exchange connectivity services. It is not possible to tell how much of the connectivity fees cover SIP connectivity and how much of them cover connectivity services for other exchange products such as proprietary data feeds. For example, one exchange stated that "users can connect to Regulation NMS equities and options feeds disseminated by the SIP using either of the co-location local area networks. Users do not pay an additional charge to connect to the NMS feeds: it comes with their connection to the local area network." See NYSE's Notice of Filing of Proposed Rule Change to Amend the Exchange's Price List Related to Co-location Services, available at https://www.sec.gov/rules/sro/nyse/2019/34-86865.pdf (last accessed Sept. 8, 2020). For this reason, the Commission's estimates include several assumptions.

2018 SIP data connectivity revenues range from approximately $13[2076] million to $18 million.[2077]

2076 Neither of the exchanges operating the exclusive SIPs disclose their connectivity revenues as a separate item on their Form 1 filings. However, one of the commenters disclosed its connectivity revenues to be $167.6 million in 2018. See Nasdaq Letter IV at 29. To estimate the portion of this connectivity revenue that comes from subscribers of SIP data, the Commission uses the ratio of non-display SIP data revenues with respect to that the exchange's overall market data revenues (excluding connectivity fees). The Commission uses a revenue ratio based on non-display SIP data revenues within the overall market data revenues, because non-display data subscribers are the most likely connectivity purchasers for SIP data. Nasdaq's information services revenues (which covers its market data revenues, excluding its connectivity revenues) at the end of 2018 were approximately $243 million. See supra note 2072 for Nasdaq's 2019 Form 1 filings. In the same time period, its total non-display SIP revenues were approximately $9 million. See infra note 2191 for the CTA and UTP Plans' Q2 2020 Quarterly Revenue Disclosure. In other words, Nasdaq's total non-display SIP data revenues were approximately 4% of its overall market data revenues, excluding the connectivity revenues. For the lower bound estimation, the Commission assumed that the other exchange operating an exclusive SIP has the same amount of connectivity revenue from SIP data as Nasdaq ($6.4 million), bringing the lower bound of the total SIP data connectivity revenues that might be available to competing consolidators to approximately $13 million ($6.4 million times 2).

2077 For the upper bound estimates, the Commission calculates the following numbers. First, the Commission estimates Nasdaq's connectivity revenues the same way, approximately $6.4 million in 2018. Second, the Commission assumed that, in 2018, NYSE had the same amount of total connectivity revenue as Nasdaq ($167.6 million). See supra note 2072 for NYSE's 2019 Form 1 filings. To estimate the portion of this connectivity revenue that comes from subscribers of SIP data, the Commission similarly used the ratio of non-display SIP data revenues with respect to that the exchange's overall market data revenues (excluding connectivity fees). In 2018, NYSE's non-display data revenue from the SIPs were approximately $5 million and its overall market data revenue for the same period (excluding the connectivity revenues) were approximately $68 million ($236 million overall data services revenues minus the $167.6 million connectivity revenues). This indicates an approximately 7% revenue ratio. The Commission, then, estimated that NYSE's SIP data connectivity revenues in 2018 were approximately $12 million (7% x $167.6 million). This brings the upper bound of the total SIP data connectivity revenues that might be available to competing consolidators to approximately $18 million ($6.4 million plus $12 million).

Finally, the Commission also believes that a number of firms may switch from using proprietary data feeds to using consolidated market data products provided by competing consolidators.[2078] The Commission believes that a reasonable range of firms who could switch to using consolidated market data products from using proprietary data feeds is 10 to 15. A typical firm using non-display feeds typically requires feeds from 10 exchanges,[2079] which the Commission estimates would cost approximately $1.1 million per year, per use case.[2080] The Commission also believes that the typical broker-dealer firm would have 2 use cases, so that the total spent on these proprietary data feeds would be $2.2 million. Using the 6% fee charged by normalizers discussed above, the Commission believes that there is between approximately $1.3 million and $2 million in revenue available to competing consolidators from this market segment.

If these 10 to 15 firms switch from using proprietary market data obtained from direct connections to the exchanges to using a competing consolidator, then they will no longer pay connectivity fees to the exchanges for their data access.[2081] As in the case of connectivity fees for the exclusive SIPs, the Commission believes that the connectivity fees for the proprietary feed connections to which these market participants cease to subscribe represents potential

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2078 See infra Section V.C.4(a) for additional discussion of this point, including how the expanded content of core data will be part of the reason firms may switch. These details were also discussed in the Proposing Release 85 FR at 16853.

2079 Specifically, those exchanges are NYSE, NYSE American, NYSE Arca, Nasdaq, Nasdaq BX, PSX, Cboe BYX, Cboe BZX, Cboe EDGA, and Cboe EDGX.

2080 See Katsuyama Letter II; Letter from John Ramsay, Chief Market Policy Officer, Investors Exchange LLC, to Brent J. Fields, Secretary, Commission (Feb. 4, 2019)

2081 See supra Section V.B.3(c) for discussion of connectivity services.

revenue for competing consolidators. The Commission estimates that the connectivity fees associated with these 10 to 15 dropped connections total approximately $27 million to $40 million.[2082]

The Commission believes that even though the $78 to $97 million estimated potential annual revenue is an underestimate, it is still large enough to support multiple competing consolidators. The estimated ongoing costs per competing consolidator range from $6.6 million to $8 million (including the ongoing Regulation SCI costs),[2083] leaving substantial room for profits for multiple competing consolidators even after incurring initial costs.

Finally, some commenters argued that most market participants interested in consolidated market data might become self-aggregators, which might shrink the customer base available to competing consolidators preventing the emergence of a healthy competing consolidator

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2082 This estimate follows a similar methodology as in note2076. The Commission assumed that a given percentage of total proprietary data feed revenue comes from customers who also make up the same percentage of proprietary data connectivity revenue. The Commission estimates that 10 to 15 proprietary data customers, each with 2 non-display use cases, represent approximately $21.9 million to $32.9 million in proprietary data revenue. Using the revenue numbers from Section V.B.2(e), the Commission estimates that this is approximately between 8% and 12% of total exchange proprietary data revenue ($21.9 million / $269.0 million; $32.9 million / $269.0 million). The assumption that these customers make up the same percentage of total exchange connectivity revenue yields that these customers are responsible for between $26.6 million and $40.0 million of total exchange connectivity revenue (8% x $327 million; 12% x $327 million). In this calculation, the connectivity revenue that pertains to the exclusive SIPs is subtracted from total connectivity revenue to produce the base of proprietary data connectivity revenue. The conservative estimate of the upper bound on SIP connectivity revenue discussed in note 2077 of $18 million was used in both cases (10 and 15 switching users) to yield $327 million for proprietary connectivity revenue.

2083 See infra Sections V.C.2(d) and V.C.2(e)(ii) for a discussion about the ongoing cost potential competing consolidators might incur. See also infra note 2256 for the total direct cost numbers.

market.[2084] While acknowledging that some market participants might become self-aggregators,[2085] the Commission believes the market will still support multiple competing consolidators. A variety of market participants will likely demand the entirety or a subset of consolidated market data, including market participants who currently rely on SIP data as well as market participants who might switch from the exchanges' proprietary data feeds to consolidated market data.[2086] However, only a small portion of these are permitted to and will likely choose to be self-aggregators. For instance, few, if any, of the market participants who currently rely only on SIP data will become self-aggregators under the amendments because of the extensive investment and technical expertise that is needed to become a self-aggregator.[2087] Additionally, of the market participants who might switch from using the exchanges' proprietary data to consolidated market data, only certain market participants and with certain limitations are permitted to self-aggregate under the amendments.[2088] Other market participants who are not permitted to self-aggregate but who are consumers of the exchanges' proprietary data will need to subscribe to a competing consolidator if they switch to using consolidated market data.[2089] On

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2084 See, e.g., NYSE Letter II at 17; Nasdaq Letter IV at 8, 24; Nasdaq Letter V at 7; STANY Letter at 7; IDS Letter I at 15.

2085 One commenter stated that they would expect some of their "members to consider becoming self-aggregators pursuant to the Proposal." See FIA PTG Letter at 1–2.

2086 See infra Section V.C.4(a) for a discussion on some market participants' potential switch to consolidated market data.

2087 See supra note 2073.

2088 See supra Section III.D for a discussion on the market participants that can be self-aggregators and the conditions under which they can share data.

2089 One commenter said that "many non-broker-dealer market participants subscribe directly to proprietary data feeds from exchanges" and that they will likely want to use consolidated market data. See MFA Letter at 3. One asset manager commented that they